

DIAMOND PHU

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EDUCATION

University of California, Santa Cruz
B.S. in Computer Science

Santa Cruz, CA
Sep. 2019 – Jul. 2022

EXPERIENCE

Software Engineer | Naval Air Systems Command (NAVAIR)

Oct. 2022 – Sept. 2025

C#, .Net, WPF, AvaloniaUI, SQL, Powershell, Microsoft Sharepoint

Point Mugu, CA

- Supported over 80 high-complexity test and evaluation events for aircraft, ship, and missile systems by troubleshooting real-time systems and enterprise networks in defense and aerospace environments.
- Architected and deployed the first t-SQL database for multiple telemetry hardware, implementing a normalized schema and optimized queries; built a full-stack C# desktop application with .NET and Avalonia UI to monitor, manage, and visualize real-time database transactions; stored and managed configurations for various scheduled real-time events to support diverse test scenarios.
- Developed C# applications using .NET, WPF, and Avalonia UI to parse 12 hours of PCAPNG network recordings (split into 1-hour files) in parallel, converting GPS, telemetry, and radar flight data into CSV and raw data formats with real-time progress visualization; executed after every real-time operation to deliver processed data products to external and internal customers accross stakeholder teams.
- Validated a new data processing application by running simulations against GPS, telemetry, and radar playback data, comparing outputs to the legacy system via graphical analysis to confirm production readiness.
- Conducted pre-deployment testing in isolated Linux environments using CLI tools (tcpdump, Wireshark) for packet capture and network traffic analysis; configured TCP/UDP data flow, firewall rules, and Cisco switch settings (IGMP snooping, port security, VLAN).
- Designed and deployed a PowerShell-based system configuration tool with a UI to automate system setup (IP address, DNS, system name) from CSV-stored configuration data; used during testing environment setup and scheduled maintenance days to apply new network configurations and deploy updated system images across 100+ mission-critical systems per functional area within the sea range facility, collaborating with stakeholders on image requirements and group policies.

PROJECTS

AI Task Manager | Python

- Built a full-stack AI productivity assistant using FastAPI (Python) and React/TypeScript.
- Integrated OpenAI API (GPT-4o-mini) with structured function calling to extract and track tasks from natural conversation.
- Implemented a multi-turn chat system with hourly check-ins, calendar event scheduling, and task checklist management to monitor meetings, deadlines, and task progress.
- Designed modular backend architecture separating schemas, agents, and route handlers following industry best practices.

Auto-Scheduler | JavaScript, React, TypeScript

- Built a full-stack single-page React/TypeScript portfolio + calendar scheduling app deployed to GitHub Pages via Vite, featuring an interactive Leaflet map with SF and Daly City street sweeping data and a monthly/hourly planner supporting daily, weekly, biweekly, and monthly recurring events plus holidays.
- Engineered a multi-source ETL pipeline (Python BeautifulSoup scraper + Node.js normalizer) to parse 38,700+ street sweeping schedule entries from HTML tables and GeoJSON into a unified dataset, with geospatial lookup and fuzzy street matching.
- Integrated Google OAuth 2.0 and the Google Calendar API to authenticate users and auto-generate recurring calendar reminders (full CRUD support) directly from map selections.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C#, SQL, PowerShell, C, C++, Java, Assembly (MIPS)

Technologies: React, .NET, WPF, AvaloniaUI, Linux/Unix, Git, SQL Server, PostgreSQL, MySQL, Docker, Virtual Machines, Pandas, NumPy, Wireshark

SECURITY CLEARANCE

Clearance Level: Secret - Current